

Offline Coefficient-Space Comparison for Case-2 Maps ANN vs GPR, $n_s \in \{131, 141, 151\}$

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Objective

This report compares six previously trained Case-2 maps in pure offline reconstruction mode:

$$(\mu, t) \mapsto q_s(\mu, t),$$

without running online nonlinear ROM solves.

The compared models include ANN and GPR maps with

$$n_s \in \{131, 141, 151\},$$

using the same linear PROM reference trajectories.

Reference trajectories are the linear PROM coefficients $q_N^{lin}(t; \mu)$ with $n_{tot} = 151$ at

$$\mu^{(v)} = (4.875, 0.0225), \quad \mu^{(1)} = (4.560, 0.0190), \quad \mu^{(2)} = (5.190, 0.0260).$$

Model specifications used in this report

All maps use inputs (μ_1, μ_2, t) . ANN131/ANN141 output q_s ; ANN151 outputs the full q_N . For ANN151, the hidden architecture is the same as in the training script core MLP:

(32, 64, 128, 256, 256) with ELU activations.

Table 1: ANN training setup (from Stage-3 summaries).

n_s	hidden dims	act.	drop.	batch	lr	weight decay	epochs ran	best val MSE
131	(32, 64, 128, 256, 256)	elu	0.0	128	0.001	1e-06	1297	0.2420964539051056
141	(32, 64, 128, 256, 256)	elu	0.0	128	0.001	1e-06	2000	0.1521771103143692
151	(32, 64, 128, 256, 256)	elu	0.0	128	1e-3	1e-6	2000	0.3817078769207001

Table 2: GPR kernels after optimization, including WhiteKernel terms.

n_s	kernel init	kernel learned	α	WhiteK	WhiteK bounds	val rel. Frobenius (%)
131	1**2 * Matern(length_scale=[1, 1, 1], nu=1.5) + WhiteKernel(noise_level=1e-10)	2.67**2 * Matern(length_scale=[0.0001, 0.0001, 0.146], nu=1.5) + WhiteKernel(noise_level=1e-10)	1e-12	True	(1e-10, 1.0)	0.9932419125450657
141	1**2 * Matern(length_scale=[1, 1, 1], nu=1.5) + WhiteKernel(noise_level=1e-10)	5.01**2 * Matern(length_scale=[0.0001, 6.39, 0.236], nu=1.5) + WhiteKernel(noise_level=1e-10)	1e-12	True	(1e-10, 1.0)	0.420604737643367
151	1**2 * Matern(length_scale=[1, 1, 1], nu=1.5) + WhiteKernel(noise_level=1e-10)	4.86**2 * Matern(length_scale=[0.0001, 6.42, 0.237], nu=1.5) + WhiteKernel(noise_level=1e-10)	1e-12	True	(1e-10, 1.0)	0.0899985346079204

Error definitions

For a model with split $n_{tot} = n_p + n_s$, we compare predicted secondary coefficients against the corresponding tail of the linear reference:

$$q_s^{ref}(t; \mu) = [q_N^{lin}(t; \mu)]_{n_p+1:n_{tot}}, \quad q_s^{pred}(t; \mu) = \mathcal{M}_\theta(\mu, t).$$

For each predicted coefficient index $i = 1, \dots, n_s$:

$$E_i^{abs}(\mu) = \|q_{s,i}^{ref} - q_{s,i}^{pred}\|_2, \quad E_i^{rel}(\mu) = 100 \frac{\|q_{s,i}^{ref} - q_{s,i}^{pred}\|_2}{\|q_{s,i}^{ref}\|_2 + \varepsilon}.$$

Global secondary-trajectory error:

$$E_F(\mu) = 100 \frac{\|Q_s^{ref} - Q_s^{pred}\|_F}{\|Q_s^{ref}\|_F + \varepsilon}.$$

Since models have different n_s , each curve starts at global index $n_p + 1$ with $n_p = n_{tot} - n_s$. Therefore, missing lower global indices for some curves are expected.

Pointwise summary

Table 3: Offline coefficient errors per point (%).

Model	n_s	E_F	$\text{mean}(E_i^{rel})$	$\text{p95}(E_i^{rel})$	$\text{max}(E_i^{rel})$
$\mu = (4.875, 0.0225)$					
ANN	131	11.020	27.578	64.427	77.394
ANN	141	4.968	17.719	40.772	51.943
ANN	151	1.791	30.688	79.297	87.480
GPR	131	0.562	1.313	4.746	5.443
GPR	141	0.162	0.669	3.060	3.842
GPR	151	0.034	0.624	2.987	3.841
$\mu = (4.560, 0.0190)$					
ANN	131	102.964	142.204	201.576	272.048
ANN	141	9.206	27.858	59.705	78.524
ANN	151	3.842	49.200	101.068	118.742
GPR	131	100.027	100.004	100.053	100.383
GPR	141	100.004	100.001	100.071	100.383
GPR	151	100.200	100.006	100.103	100.408
$\mu = (5.190, 0.0260)$					
ANN	131	69.570	113.381	170.202	201.128
ANN	141	10.268	33.535	64.266	80.333
ANN	151	5.474	53.547	105.780	138.958
GPR	131	99.987	100.003	100.046	100.083
GPR	141	99.998	100.005	100.057	100.171
GPR	151	99.892	100.003	100.061	100.171

Average over the three points

Table 4: Averaged offline coefficient errors over the three points (%).

Model	n_s	mean E_F	mean(mean(E_i^{rel}))	mean(p95(E_i^{rel}))	mean(max(E_i^{rel}))
ANN	131	61.185	94.387	145.402	183.523
ANN	141	8.147	26.371	54.914	70.267
ANN	151	3.702	44.479	95.382	115.060
GPR	131	66.859	67.107	68.282	68.636
GPR	141	66.721	66.892	67.729	68.132
GPR	151	66.709	66.878	67.717	68.140

Per-coefficient curves

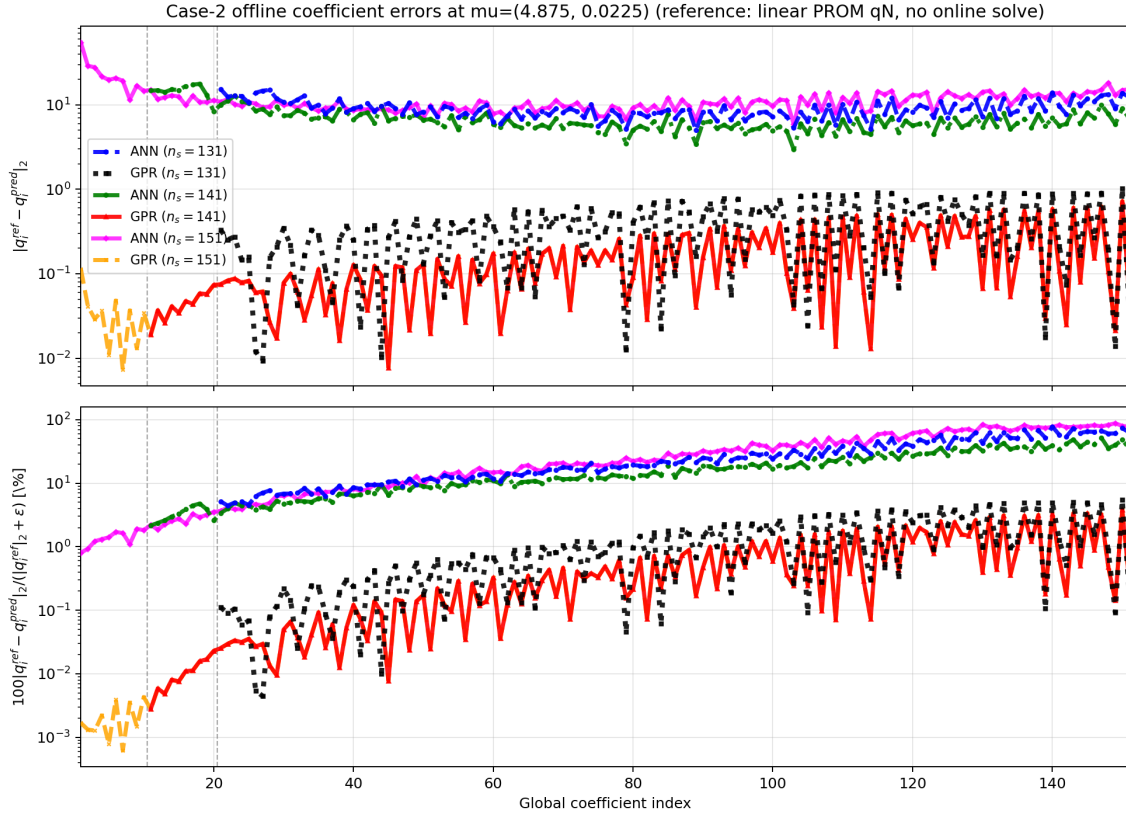


Figure 1: Absolute and relative coefficient errors vs global coefficient index at $\mu^{(v)} = (4.875, 0.0225)$.

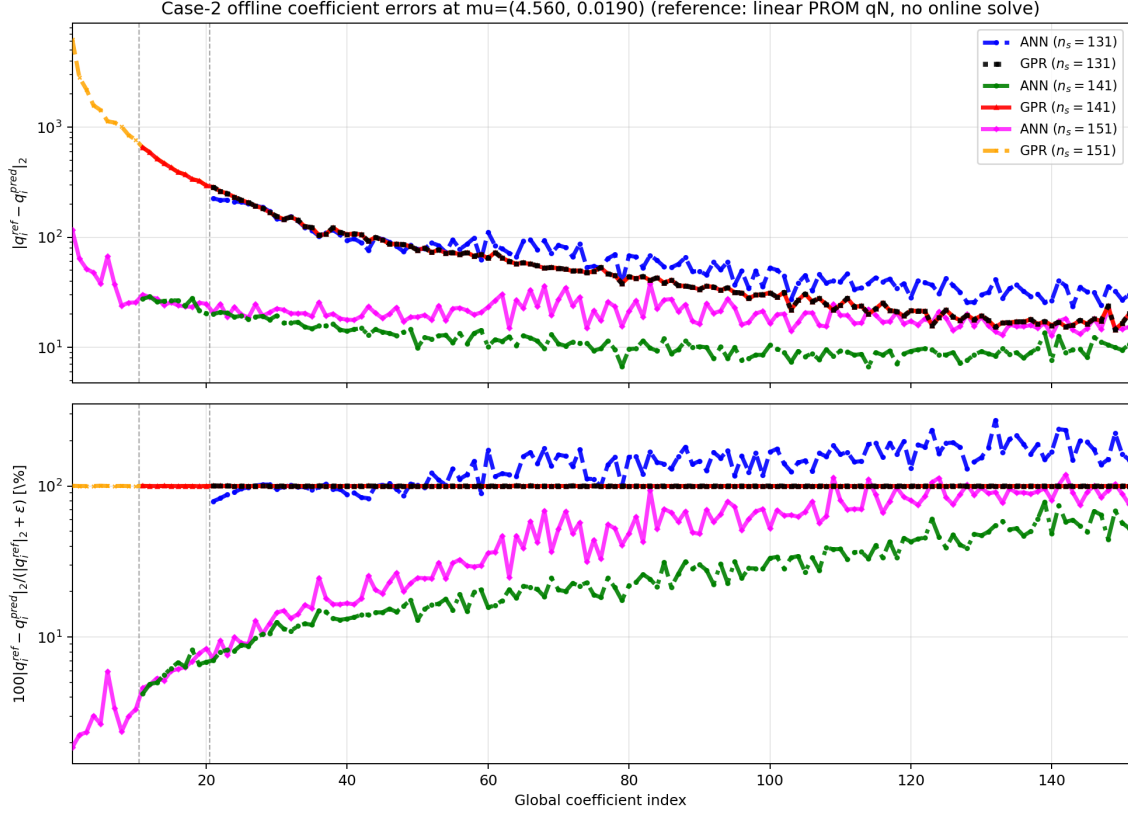


Figure 2: Absolute and relative coefficient errors vs global coefficient index at $\mu^{(1)} = (4.560, 0.0190)$.

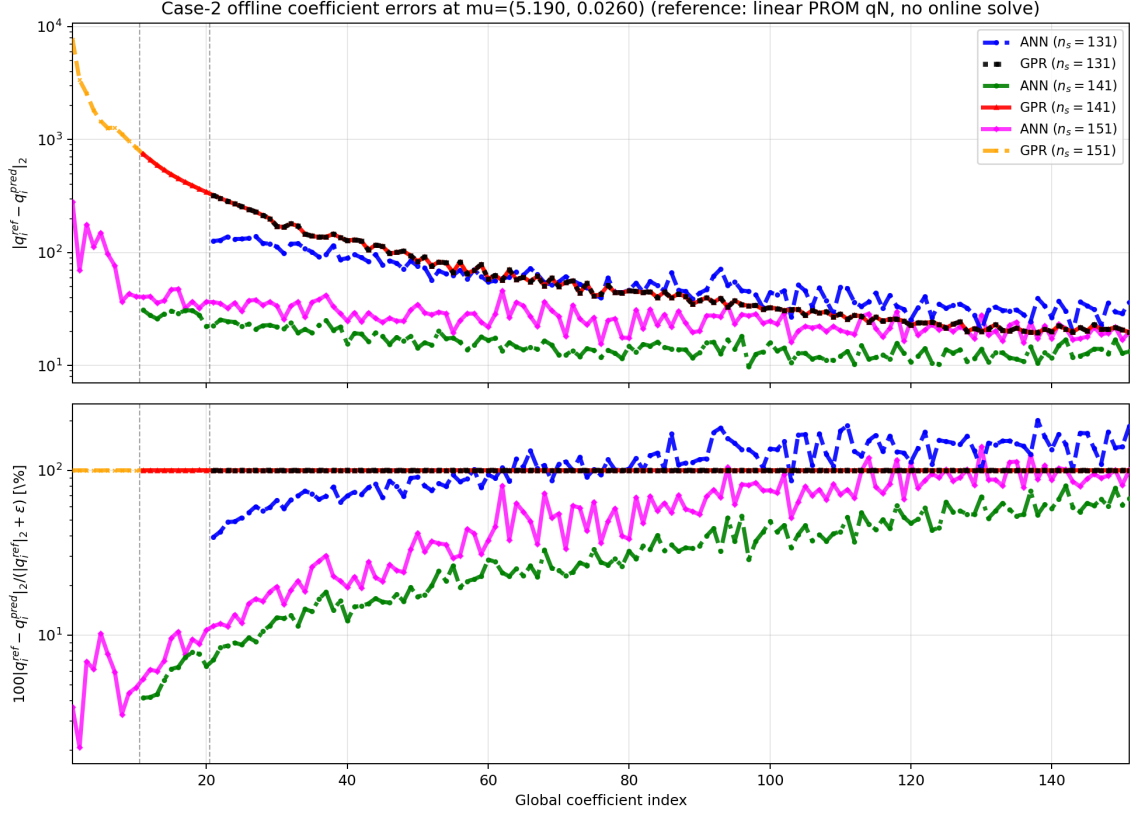


Figure 3: Absolute and relative coefficient errors vs global coefficient index at $\mu^{(2)} = (5.190, 0.0260)$.

GPR zoom

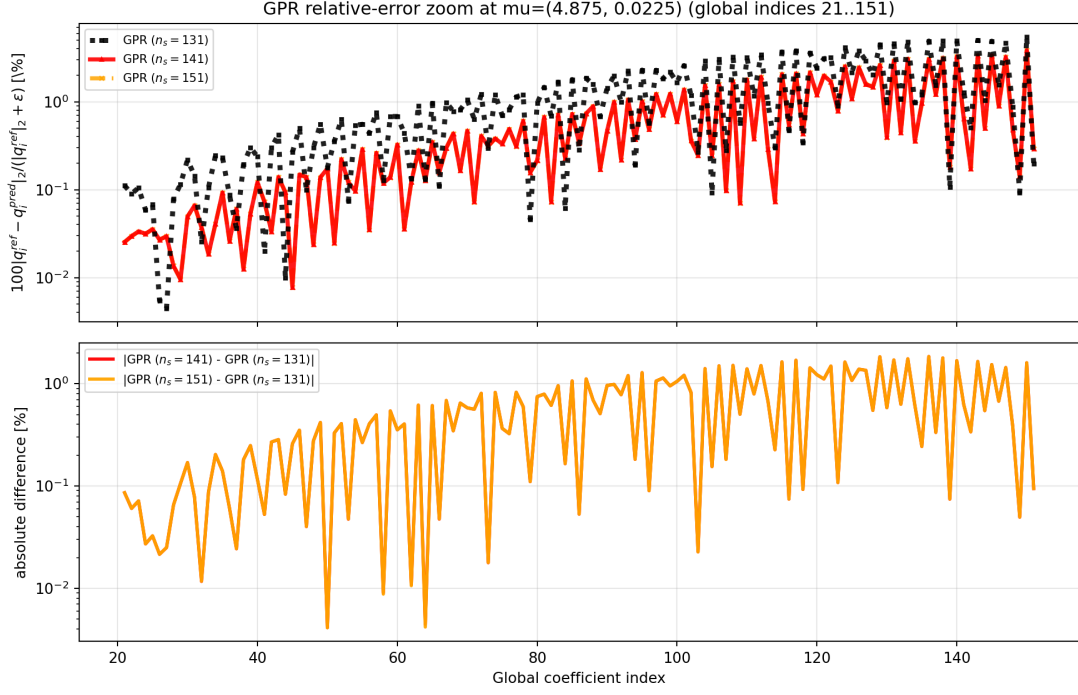


Figure 4: Zoomed GPR comparison at $\mu^{(v)} = (4.875, 0.0225)$, including pairwise differences.

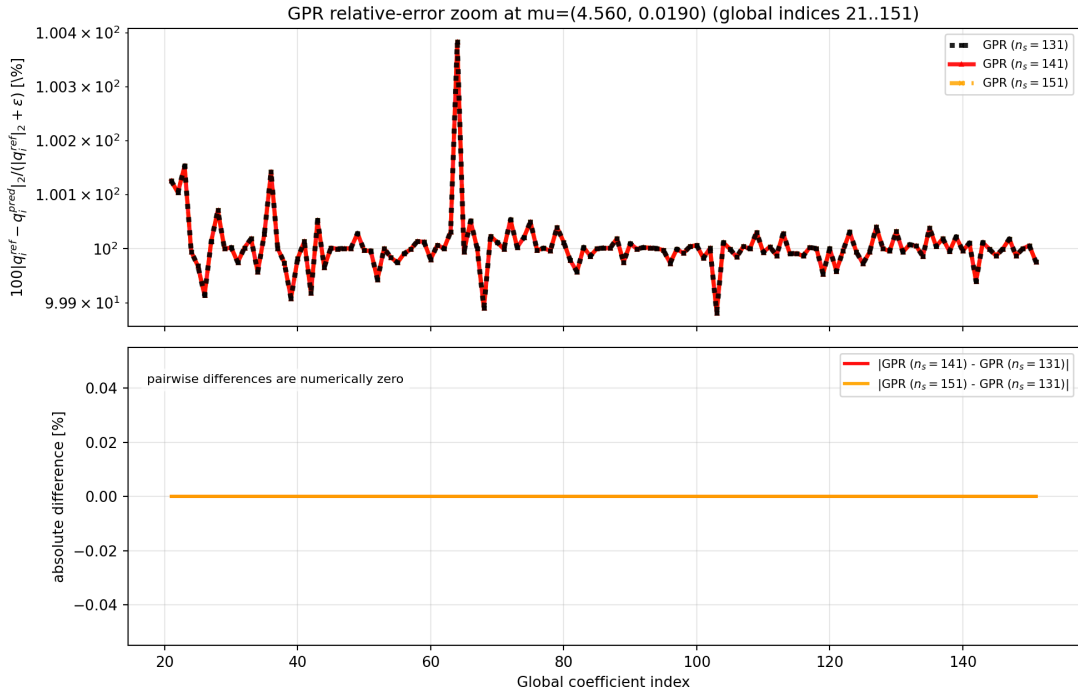


Figure 5: Zoomed GPR comparison at $\mu^{(1)} = (4.560, 0.0190)$, including pairwise differences.

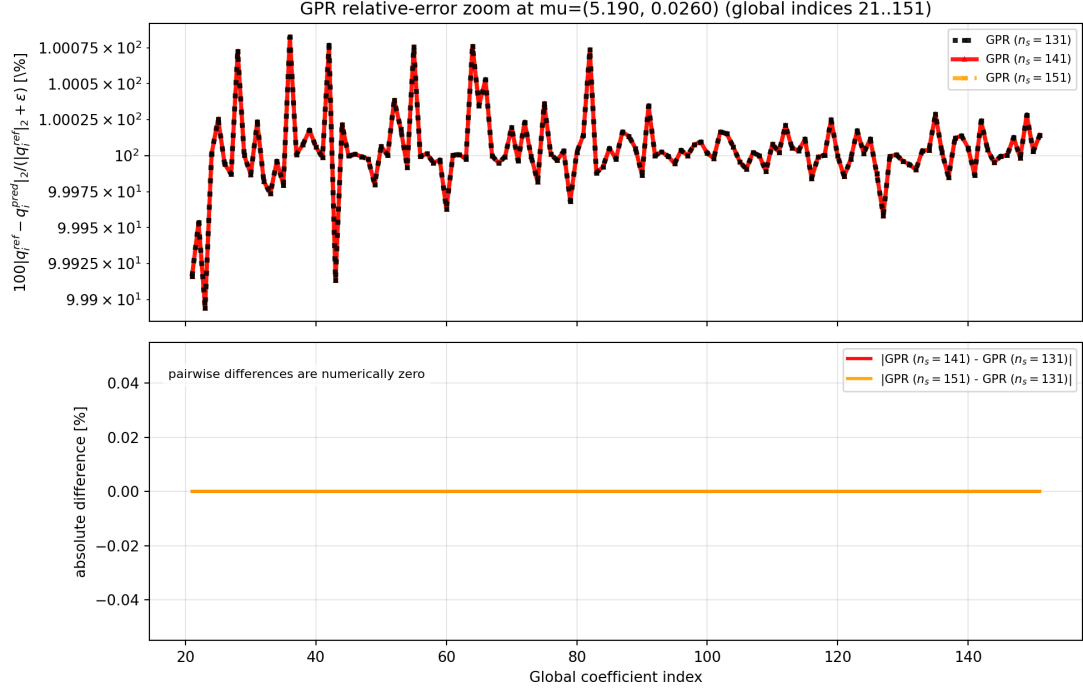


Figure 6: Zoomed GPR comparison at $\mu^{(2)} = (5.190, 0.0260)$, including pairwise differences.

Hybrid state-space overlays (HDM comparison)

Using the same predicted coefficients, hybrid full reduced trajectories were built as

$$q_N^{hyb} = \begin{bmatrix} q_p^{lin} \\ q_s^{pred} \end{bmatrix},$$

with $(n_p, n_s) = (20, 131)$, $(10, 141)$, and $(0, 151)$ depending on the model. Then states were reconstructed by

$$u^{hyb}(t) = u_{ref} + V_{tot} q_N^{hyb}(t),$$

and compared directly against HDM snapshots.

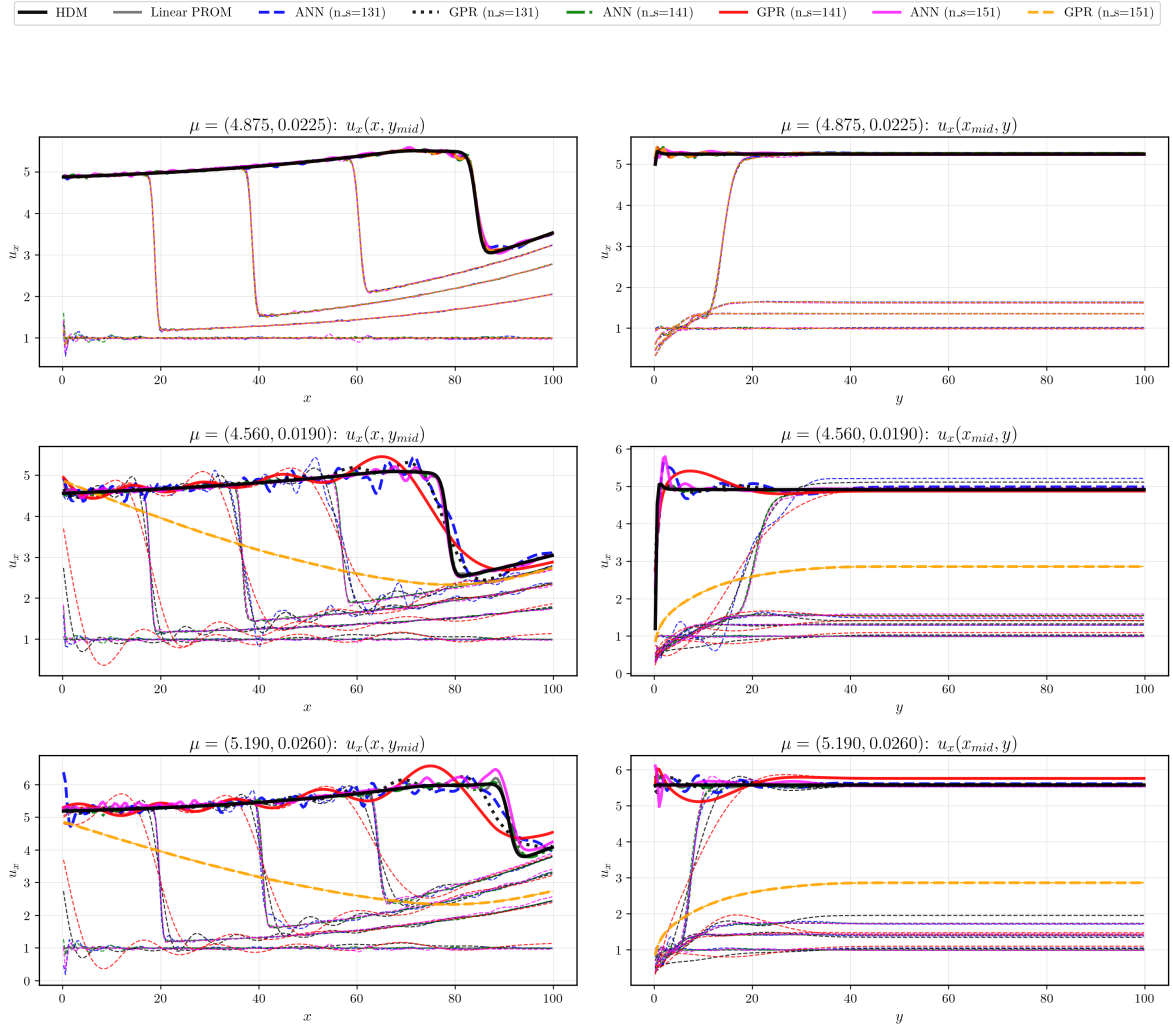
Table 5: Hybrid state errors per point (%).

Model	n_s	err. vs HDM (%)	err. vs linear state (%)
$\mu = (4.875, 0.0225)$			
ANN	131	0.633	0.535
ANN	141	0.599	0.447
ANN	151	0.849	0.769
GPR	131	0.411	0.027
GPR	141	0.411	0.015
GPR	151	0.411	0.015
$\mu = (4.560, 0.0190)$			
ANN	131	5.300	5.245
ANN	141	0.954	0.856
ANN	151	1.714	1.669
GPR	131	5.147	5.095
GPR	141	9.342	9.295
GPR	151	43.544	43.525
$\mu = (5.190, 0.0260)$			
ANN	131	3.448	3.424
ANN	141	1.014	0.919
ANN	151	2.441	2.405
GPR	131	4.926	4.921
GPR	141	8.944	8.947
GPR	151	43.879	43.886

Table 6: Average hybrid state errors over the three points (%).

Model	n_s	mean err. vs HDM (%)	mean err. vs linear state (%)
ANN	131	3.127	3.068
ANN	141	0.856	0.741
ANN	151	1.668	1.614
GPR	131	3.495	3.348
GPR	141	6.232	6.085
GPR	151	29.278	29.142

Offline hybrid state overlays: HDM vs ANN/GPR hybrid qN reconstructions



Dashed: intermediate times; solid: final time $t = 25$. Hybrid rule: low block from linear PROM, high block from map.

Figure 7: Hybrid offline state overlays: HDM, linear PROM reference, and ANN/GPR hybrid reconstructions for $n_s \in \{131, 141, 151\}$.

Remarks

This report is strictly an offline map-comparison diagnostic in coefficient space. It does not include online ROM reconstruction of state trajectories.